

Dissertation / Project / Project Work Title:
AI Inference Acceleration At Edge Using Neural Processing Unit

Course No. : S2-25_ESZG628T

Course Title: Dissertation

Dissertation / Project /Project Work Done by:

Student Name: Bhimesh Kiritkumar Bhatt

BITS ID: 2024HT01120

Degree Program: M. Tech. Embedded Systems

**Research Area: AI Workload Acceleration At Edge
Using NPU**

Dissertation / Project Work carried out at: BITS Pilani



**BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE,
PILANI
VIDYA VIHAR, PILANI, RAJASTHAN -
333031.**

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Contents

1. Broad Area of Work	3
2. Background	3
3. Objectives	3
4. Scope of Work	4
5. Plan of Work	4
6. Literature References	5
7. Particulars of the Supervisor and Examiner	6
8. Remarks of the Supervisor	6

1. Broad Area of Work

The explosion of smart devices and the demand for real-time intelligence have exposed the limitations of traditional, cloud-centric AI inferencing. Sending massive data in terms of observations, pictures, voice and video streams to the cloud for processing is no longer a viable solution. The challenges are significant in terms of Latency, Bandwidth, Cost, Privacy & Security and Reliability.

AI inferencing at Edge is the paradigm shift that addresses these challenges head-on. By processing data directly on the device we can unlock a new realm of possibilities for real-time intelligent application.

My project work focuses on studying the role of various processing units such as CPU, GPU, NPU, TPU etc. (Hybrid SoC) in accelerating the AI inferencing at Edge. My project also focuses on selecting suitable AI algorithm for object detection and its optimization techniques for implementing it on resource constrained devices.

2. Background

Tata Consultancy Services offers comprehensive AI at edge solutions designed to bring intelligence closure to the source of data – such as IOT devices, machines and autonomous vehicles – to reduce latency, enhance privacy and enable real time decision making. These service are built around AI powered computer vision, cognitive analytics and IOT integration. TCS leverages partnership with leaders like Qualcomm, NVIDIA, AWS and Microsoft for researching, developing and implementing AI solutions. One of the core focus area of TCS is to build solutions for analytics using AI at edge such as computer vision, image / video analytics for quality inspection, anomaly detection, worker safety and enhanced safety in vehicles.

This project focuses on one of the key area of providing enhanced safety in vehicles leveraging computer vision at edge and will help provide direction in building new solutions and identifying new research areas related to low cost, real time systems, AI algorithms and data requirements for future work.

3. Objectives

The objectives of my project are as follows:

- Study microcontroller based architectures of various processing units such as CPU, GPU and NPU and its role in accelerating AI inference at Edge.

- Study common CNN based algorithms for Real-time Object Detection at Edge such as Yolo.
- Study the Algorithm Optimization Techniques to run AI algorithm on resource constrained microcontroller based systems.
- Study the partitioning of AI algorithm between various XPU's to improve performance of the system.
- Build an Object Detection system to achieve above objectives

4. Scope of Work

- Study, compare and contrast architectures of CPU, GPU, NPU and TPU for executing AI workload at Edge.
- For practical implementation, study architecture of STM32N6570 micro-controller which is based on ARM Cortex M55 CPU architecture and Neural-Art Accelerator for AI which is a NPU architecture from ST Microelectronics
- Study popular AI algorithm like Yolo and its various implementations for object detection and select the one suitable for our system.
- Select readily available AI Model based on Yolo for detection of common objects on road. Train the model if such available model is not sufficient by using readily available data set or create a data set if not sufficient data set is available.
- Design a Driver Assistance system using STM32N6570 Discovery kit (or using Single Board Computer like Rpi and NPU accelerator) to notify driver about common objects on road such as Signal, Bumper, Zebra Crossing, Vehicle ahead, People crossing roads and common road signs (like speed limit, intersection, dangerous bend etc.)

5. Plan of Work

Phases	Start Date-End Date	Work to be done
Dissertation Outline	1 Jan 2026 – 30 Jan 2026	Literature Review and prepare Dissertation Outline
Check Point	31 Jan 2026	Outline Report Submission
Design & Development	1 Feb 2026 – 20 Feb 2026	- Literature Review (XPU architectures, CNN based Yolo Algorithms, STM32N6570 Architecture) - Parts procurement

		- Report Preparation
	21 Feb 2026 – 20 Mar 2026	- Design, Data Gathering, Model Training, Model Quantization, Model implementation and optimization - Sandbox testing - Report Preparation
Check Point	21 Mar 2026	Mid Sem Report Submission
	22 Mar 2026 – 10 Apr 2026	- Implement the AI Model on target system. - Fine tune model implementation - Performance Measurements and compare various implementation options
Testing	11 Apr 2026 – 20 Apr 2026	- Testing in real life setting - Fine tuning - Final Report Preparation
Dissertation Review	20 Apr 2026 – 30 Apr 2026	Submit Dissertation to Supervisor & Additional Examiner for review and feedback
Submission	1 May 2026 – 12 May 2026	Final Review and submission of Dissertation

6. Literature References

Following are the key references for this work.

STM32N6570 Discovery Kit

<https://www.st.com/en/evaluation-tools/stm32n6570-dk.html>

Yolo Documentation

<https://docs.ultralytics.com/>

Fast Object Detection with Machine Learning Edge Device

https://www.researchgate.net/publication/384698711_Fast_Object_Detection_with_a_Machine_Learning_Edge_Device

Hardware Acceleration for Neural Networks: A Comprehensive Survey

[Bin Xu](#), [Ayan Banerjee](#), [Sandeep Gupta](#)

<https://arxiv.org/abs/2512.23914>

Neural Network Quantization for Microcontrollers: A Comprehensive Survey of Methods, Platforms, and Applications

[Hamza A. Abushahla](#), [Dara Varam](#), [Ariel Justine N. Panopio](#), [Mohamed I. AlHajri](#)

<https://arxiv.org/abs/2508.15008>

Benchmarking Energy and Latency in TinyML: A Novel Method for Resource-Constrained AI

[Pietro Bartoli](#), [Christian Veronesi](#), [Andrea Giudici](#), [David Siorpaes](#), [Diana Trojaniello](#), [Franco Zappa](#)

<https://arxiv.org/abs/2505.15622>

ActNAS : Generating Efficient YOLO Models using Activation NAS

[Sudhakar Sah](#), [Ravish Kumar](#), [Darshan C. Ganji](#), [Ehsan Saboori](#)

<https://arxiv.org/abs/2410.10887>

Comparative Analysis of Deep Learning Models for Perception in Autonomous Vehicles

[Jalal Khan](#)

<https://arxiv.org/abs/2512.21673>

7. Particulars of the Supervisor and Examiner

	Supervisor	Additional Examiner
Name	Gaurav Rewari	Apoorve Aggarwal
Qualification	B. Tech.	B. Tech.
Designation	Principle Consultant	Senior Consultant
Employing Organization and Location	TCS, New Delhi	TC, New Delhi
Phone No. (with STD Code)	+91 9871172145	+91 9891909960
Email Address	Gaurav.rewari@tcs.com	Apoorve.aggarwal@tcs.com

8. Remarks of the Supervisor

The proposed study and project are highly promising and align well with the current technological and research direction of TCS. Bhimesh Bhatt has conducted a comprehensive and relevant literature review, demonstrating a clear understanding of existing work in the domain. The project addresses a contemporary research area and adequately justifies the academic depth and rigor expected of a master's-level dissertation.

The outcomes of the project are insightful and incorporate state-of-the-art techniques relevant to today's Physical AI landscape. Additionally, the project includes a functional working prototype, which effectively validates the proposed research approach and outcomes.

The work also shows strong potential for future publication in reputed journals or conferences. Based on the initial interactions with Bhimesh Bhatt, I found him to be diligent, motivated, and technically competent, with the necessary skills to execute experiments following standard research methodologies.

In view of the above, and considering its relevance to ongoing industry and research trends, I approve the project in the capacity of supervisor.

BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI
WORK INTEGRATED LEARNING PROGRAMMES (WILP) DIVISION
SECOND SEMESTER OF ACADEMIC YEAR 2021-2022

S2-25 ESZG628T : AI Inference Acceleration At Edge Using Neural Processing Unit OUTLINE

STUDENT ID No.	2024HT01120
NAME OF THE STUDENT	Bhimesh KiritKumar Bhatt
STUDENT'S EMAIL ADDRESS	2024HT01120@wilp.bits-pilani.ac.in
STUDENT'S EMPLOYING ORGANIZATION & LOCATION	TCS, Mumbai
SUPERVISOR'S NAME	Gaurav Rewari
SUPERVISOR'S EMPLOYING ORGANIZATION & LOCATION	TCS, New Delhi
SUPERVISOR'S EMAIL ADDRESS	Gaurav.rewari@gmail.com
ADDITIONAL EXAMINAER'S NAME	Apoorve Aggarwal
ADDITIONAL EXAMINER'S EMPLOYING ORGANIZATION & LOCATION	TCS, New Delhi
ADDITIONAL EXAMINER'S EMAIL ADDRESS	Apoorve.aggarwal@tcs.com
DISSERTATION / PROJECT / PROJECT WORK TITLE	AI Inference Acceleration At Edge Using Neural Processing Unit

Please prepare the outline as a separate document with the following sections along with the above identification information.

1. Cover Page with Student ID No., Name, Course Number, Course Title and Dissertation / Project / Project Work Title, Broad Academic Area of Work.
2. Background (Relevance of the Project to the current work environment in the employing org.)
3. Objectives
4. Scope of Work (To be done by the student independently)
5. Plan of Work (Work to be done during the semester)
6. Literature References
7. Particulars of the Supervisor and Additional Examiner
8. Remarks of the Supervisor

Signature of Student	Signature of Supervisor	Signature of Additional Examiner
Name: Bhimesh Bhatt	Name: Gaurav Rewari	Name: Apoorve Aggarwal

